Semi-Automated Section in situ Hybridisation for 7μm paraffin sections

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This protocol is a compilation of protocols supplied by Jing Yu from Andy McMahon's laboratory and protocols by Wilkinson & Nieto Methods Enzymol. 1993;225:361-73. The Tecan script was supplied by Tecan and Gregor Eichele Dev Dyn. 2005 Oct;234(2):371-86 but has been modified for paraffin sections by Bree Rumballe.

Day 1:
Dewaxing of slides
Assembly of hybridisation chambers
Pre-hybridisation
Overnight hybridisation with probe
Stringency washes
Blocking
anti-DIG-AP detection

All of the solutions for Day 1 are made with RO (18ohm) -treated pure water in sterile plastic vials/glassware baked at 180°C.

Day 2:
Colour development

DAY 1:

DE-WAXING

Ensure RNase-free conditions and solutions at all times.
Label slides with pencil and assemble into baked steel racks.

In the fume hood, assemble baked glass staining jars and label with appropriate solution name. Make up 300ml of each solution. Take the slides through the following de-waxing process at room temperature:

- Immerse in Xylene 1 for 10 minutes (min).
- Immerse in Xylene 2 for 10min.
- Rehydrate in 100% ethanol for 1min.
- Rehydrate in 95% ethanol for 1min.
- Rehydrate in 80% ethanol for 1min.
- Rehydrate in 60% ethanol for 1 min.
- Rehydrate in 30% ethanol for 1 min.
- Wash in phosphate buffered saline (PBS) for 5min.
- Fix in cold freshly made 4% paraformaldehyde (PFA) for 10min.
- Wash in PBS for 5min.
- Wash in second PBS for 5min.
- Transfer to fresh PBS to assemble slides.

Assemble slides into flow-through hybridisation chambers under PBS and ensure no bubbles are present on the slides. Place into degassed chamber racks on the Tecan Freedom Evo platform.

Ensure all reagents are in the troughs in the correct location on the Tecan platform.

TREATMENT

- Begin the Tecan script by incubating slides in PBS once for 5min at 25°C.
- Incubate twice for 5-10min with 10µg/ml proteinase K (Roche 3115879) in TBS. Proteinase K is freshly added to the buffer as the script commences, mixed and placed onto the Tecan platform. Adjustment of proteinase K concentration/incubation time may be required for different developmental stages and tissue types. Try 10min for embryonic, 20min for adult tissue.
- Incubate three times for 3min with PBS.
- Fix with 4%paraformaldehyde once for 5min.
- Wash three times for 3min with PBS.
- Incubate 5 times for 2min with 0.1M triethanolamine,0.65%HCl and 0.25% (v/v) acetic anhydride (add fresh) in water.
- Incubate three times for 5min with PBS.
- Incubate once for 3min with 0.85% NaCl.
- Wash with 70% ethanol/0.85%NaCl for 5min.
- Wash with 96% ethanol for 5 min.

- Incubate twice with hybridisation buffer (50% Formamide, 2xSSC pH5, 1x Denhardt's, 10% Dextran sulphate, 0.2mg/ml tRNA, 0.5mg/ml salmon sperm DNA) at which time the temperature of the chamber rack is increased to 65°C.

At this time the slides may incubate for as long as necessary until the chamber rack is at temperature and the probes are ready, generally between 10-20mins.

HYBRIDISATION

- RNA riboprobe (0.3-0.5µg/ml) is mixed with hybridisation buffer, heated at 80°C for 5min, put on ice for 5min, spun and placed on the Tecan platform. A 200µl aliquot is added to each chamber by the Tecan script and 4-5 hours later, a second aliquot of probe is added to the chamber and the hybridisation process is continued for another 4-5hours at 65°C.

STRINGENCY WASHES

Bottles containing stringency washes are preheated in the incubator at 64ºC and placed into the glass troughs on the Tecan platform. The solutions are maintained at 64°C using a script-controlled circulation bath connected to the chamber.

- Incubate once for 5min with 5xSSC (500mM NaCl, 50mM sodium citrate).
- Incubate three times for 10min with 50% Formamide, 1xSSC.
- The script pauses here to ensure you are ready to continue. Incubate twice for 5min with TNE (10mM Tris-HCl, pH7.5, 500mM NaCl, 1mM EDTA). During this time the script ramps down the chamber rack temperature to 37ºC.
- The script pauses to allow manual dispensing of RNase A (2µg/ml) in TNE, twice for 7mins at 37ºC.
- Proceed with the script for incubation with TNE at 37ºC twice for 7mins.
- Incubate twice for 10min with 2xSSC at 64ºC.
- Incubate four times for 10min with 0.2xSSC at 64ºC.

RIBOPROBE DETECTION

- Wash with 1x MBST (100mM Maleic acid, 150mM NaCl, 0.1% Tween-20, pH7.5) 3 times for 5min at 25ºC.
- Incubate 3 times for 20min with Blocking Buffer (20%heat inactivated sheep serum, 2% Blocking Reagent in 1x MBST) Roche11096176001)
- Incubate 3 times for 120min with 1/4000 anti-DIG-AP in 1%sheep serum, 2%Blocking Reagent in 1xMBST at 4ºC.
- The script pauses until you are ready to begin the final steps on Day2.

DAY 2

COLOUR DEVELOPMENT

- Wash slides in 1xMBST three times for 5min.
- Incubate 2 times for 5min with NTMT (0.1M NaCl, 0.1M Tris-HCl pH9.5, 50mM $MgCl_2$, 0.1% Tween-20) + 2mM Levamisole
- Disassemble hybridisation chambers while they are immersed in fresh NTMT, dry the edges and place into humidified trays ready for colour development
- Aliquot 200µl of BM Purple (Roche11442074001) directly onto slide. Cover lid with alfoil to create a dark environment and monitor colour reaction.
- Once staining is sufficient, wash with PBS for 5min.
- Fix in 4% parafomaldehyde for 10min.
- Wash twice for 5min with PBS.
- Mount and coverslip with aqueous mounting media.

SOLUTIONS for S-ISH using the Tecan Freedom Evo

Xylene
Use RNase-Free (RF) Xylene for both first and second wash. You may re-use the xylene for 4 runs only, provided it has been stored in RF bottles.
100%, 95%, 80%, 60%, 30% ethanol
300ml are required for each staining jar, use RF 100% ethanol and RO water
95% EtOH = 285ml EtOH + 15ml H_2O
80% EtOH = 240ml EtOH + 60ml H_2O
60% EtOH = 180ml EtOH + 120ml H20
30% EtOH = 90ml EtOH + 210ml H20

10xPBS (phosphate buffered saline) Stock solution
Dissolve 10 RF PBS tablets in 1L RO-H_2O. Autoclave
For 1L of 1xPBS solution, use 100ml of 10xPBS + 900ml RO-H_2O

4% PFA (paraformaldehyde) solution
Make fresh. Weigh 8g PFA, dissolve in 200ml RF 1x PBS in a rotating hybaid chamber at 65C or until dissolved. Keep on ice until ready to use.

Proteinase K Roche3115879 (10 µg/ml) in PBS

Use 25µl of a 20mg/ml stock solution (stored in aliquots at -20°C and mix into 50ml 1xPBS. Make fresh for each run. Proteinase K activity should be tested for each new batch.

Acetylation Solution
Add 1.33ml triethanolamine, 0.175ml 37% HCl in 100ml H20 and mix until dissolved.
Add 0.375ml acetic anhydride per 100ml immediately before use.

20xSSC stock, pH7
Dissolve 175.3gNaCl and 88.2g sodium citrate-2H2O in 800ml H2O.
Adjust pH to 7.0 with HCl and adjust vol to 1L with water.
Autoclave and store at RT.

For 200ml solutions,

2xSSC	20ml 20xSSC, 180ml water
1xSSC	10ml 20xSSC, 190ml water
0.5xSSC	5ml 20xSSC, 195ml water
0.2xSSC	2ml 20xSSC, 198ml water

Hybridisation Buffer
For 50ml,

25ml	50% Formamide
5ml 20xSSC	2x SSC, pH5
1ml 50x	1x Denhardt
10ml 50%	10% Dextran sulfate
1ml 50x	0.2mg/ml tRNA
1ml 50x	0.5mg/ml ssDNA
Xml	RO-H2O to make up to 50ml total volume

50% dextran sulphate stock
Dissolve 50g dextran sulphate in 100ml RO-H2O. Add powder a little at a time to the water on a heated magnetic stirrer to avoid clumping. Once dissolved, aliquot and store at -20°C.

50x tRNA stock
Dissolve 200mg in 20ml RO-H2O and rotate in hyb oven at 55-65degC O/N.
Aliquot into 1ml tubes and store at -20°C.

50x salmon sperm DNA stock
Mix 500mg in 10ml RO-H2O, shear with a 17 gauge needle and adjust volume to 20ml.

Aliquot into 1ml tubes and store at -20°C.

Stringency Wash Buffer 1 (50%F, 1xSSC)

For 500ml, 250ml formamide + 25ml 20xSSC+ 225ml RO-H2O

TNE (10mM Tris oH7.5, 500mM NaCl, 1mM EDTA)

For 1L, mix
10ml 1M Tris pH7.5
100ml 5M NaCl
2ml 0.5M EDTA
Remainder RO-H20

10xMBST stock (1M Maleic acid, 1.5M NaCl, no Tween-20)

Make 1L, pH with NaOH to pH7.5
To make 1xMBST, dilute 100ml 10xMBST in 1000ml RO-H20, add 1ml of 10% Tween-20

Roche Blocking Reagent Roche11096176001

For 100ml, add
2g of blocking reagent in 1xMBST. Place on heat stirrer at 65C for at least 60min.

Blocking Buffer (20% HISS, 2%Blocking Reagent/MBST)

For 50ml. mix
10ml heat inactivated sheep serum
40ml 2%BR in 1xMBST

Fab anti-DIG (Roche11093274910) in Blocking Buffer,

For 50ml, add 1/4000 = 12.5ul anti-DIG-Ab to 49.5ml 2%BR/MBST and 0.5ml 20% HISS.

NTMT (0.1M NaCl, 0.1M Tris pH9.6, 50mM MgCl2, 0.1% Tween20 w/v)

For 100mL of 1xNTMT,
5ml 1M MgCl2
1ml 10% Tween20
3.33ml 3M NaCl
10ml 1M Tris pH9.6
80.67ml RO-H2O
2mM Levamisole
 = 240.76MW x 0.002M x 0.050L
= 0.0240g in 50mL NTMT. Make up in fumehood.

BM Purple, Roche11442074001 RT, dark
Use undiluted BM Purple and directly aliquot onto slides

Associated publications

Rumballe BA, Chiu HS, Georgas KM and Little MH.
Use of in situ hybridization to examine gene expression in the embryonic, neonatal and adult urogenital system.
Methods in Molecular Biology: Kidney Development (commissioned book chapter).

Rumballe B, Georgas K, Little MH.
High-throughput paraffin section in situ hybridisation and dual immunohistochemistry
CSH Protocols, 2008

Georgas K, Rumballe B, Wilkinson L, Chiu HS, Lesieur E, Gilbert T, Little MH.
Use of dual section mRNA in situ hybridisation/immunohistochemistry to clarify gene expression patterns during the early stages of nephron development in the embryo and in the mature nephron of the adult mouse kidney.
Histochem Cell Biol. 2008 Nov;130(5):927-42. Epub 2008 Jul 11.